

Backpaper Exam - Numerical Computing

B. Math I

07 June, 2024

- (i) Duration of the exam is 3 hours.
- (ii) The maximum number of points you can score in the exam is 100.
- (iii) You are not allowed to consult any notes or external sources for the exam.

Name: _____

Roll Number: _____

1. Write down the output of the following commands in Octave:

- (a) (2 points) `polyval([1 -3 2], [2 3])`
- (b) (4 points) `(ones(2,4)-eye(2,4))*diag(2:5)*(ones(4,2)-eye(4,2))`
- (c) (4 points) `A=[1 2;3 4];B=[1 0;0 1]; disp(A.*B); disp(A*B)`

Total for Question 1: 10

2. (15 points) Let $f : [0, 1] \rightarrow [0, 1]$ be a continuously differentiable function such that $|f'(x)| < \frac{1}{2}$ for all $x \in [0, 1]$. Prove (from first principles) that for any $x_0 \in [0, 1]$, the sequence defined inductively as $x_{n+1} = f(x_n)$, $n = 1, 2, \dots$, converges to a fixed point of f .

Total for Question 2: 15

3. (a) (10 points) The *divide and average* method, an old-time method for approximating the square root of any positive number a , can be formulated via the iterative

sequence,

$$x_{n+1} = \frac{1}{2} \left(x_n + \frac{a}{x_n} \right).$$

Show that this formula is based on the Newton-Raphson method, and converges for any initial value $x_0 > 0$.

- (b) (15 points) A calculator is defective: it can only add, subtract, and multiply. Let $a > 0$. Use the Newton-Raphson Method, and the defective calculator to devise an algorithm to compute $1/a$ correct to a desired precision level. (Hint: Solve $x - \frac{1}{a} = 0$)

Total for Question 3: 25

4. (15 points) Let A be an invertible matrix in $M_{202}(\mathbb{R})$ such that $\|A\| = 100$ and $\|A\|_F = 101$ (where $\|\cdot\|_F$ denotes the Frobenius norm.) With justification, find the sharpest lower bound for the condition number of A (defined as $\kappa(A) = \|A\| \cdot \|A^{-1}\|$.)

Total for Question 4: 15

5. (10 points) Consider a function $f : \mathbb{R} \rightarrow \mathbb{R}$, and let $x_0, x_1, \dots, x_n$ be distinct points in $\mathbb{R}$. Prove the following formula for the n th divided difference,

$$f[x_0, x_1, \dots, x_n] = \sum_{i=0}^n \frac{f(x_i)}{w'(x_i)},$$

where $w(x) = (x - x_0)(x - x_1) \cdots (x - x_n)$.

Total for Question 5: 10

6. (15 points) Below is a list of the first 4 Legendre polynomials.

n	$P_n(x)$
0	1
1	x
2	$\frac{1}{2}(3x^2 - 1)$
3	$\frac{1}{5}(5x^3 - 3x)$

Compute the weights for the Gaussian quadrature method for 2 nodes and use it to write a formula for a rule of numerical integration to approximate $\int_{-1}^1 f(x) dx$ for a function f .

Total for Question 6: 15

7. (10 points) Manually perform three steps of Euler's method to solve:

$$\frac{dy}{dt} = \frac{1}{t + y + 1}, y(0) = 0,$$

with $h = 0.2$.

Total for Question 7: 10